

“SmartAid360” – A Global Disaster Response Platform

Your team has been tasked to architect **SmartAid360**, a cloud-based disaster response and humanitarian aid coordination platform. The application will be used **worldwide** during natural disasters (earthquakes, floods, epidemics, etc.) and must serve a wide range of users in real time — from **field volunteers using mobile phones** in remote areas with poor internet, to **government officials, donors, and NGO headquarters** accessing dashboards and analytics in real time.

Core Functional Requirements:

1. **Live Crisis Reporting:** Field agents must report incidents (e.g., medical emergencies, infrastructure collapse) from mobile devices — ideally with offline support and sync later.
2. **Aid Resource Allocation Engine:** Smart algorithms must match needs with available resources (tents, food, medicine) across regions.
3. **Real-Time Dashboard:** Admins and donors expect a live dashboard showing crisis heat maps, fund disbursement, supply chains, and impact metrics.
4. **Donation Processing:** Accept international donations via credit card, crypto, and local payment gateways, ensuring fraud protection.
5. **Audit & Compliance:** The system must meet regulatory requirements in multiple countries, including GDPR, HIPAA, and financial audits.
6. **Plug-in Model:** Partner NGOs and governments can plug in their own apps or modules.
7. **AI/ML Forecasting Engine:** Predict future crises or resource shortfalls based on real-time and historical data.

Additional Context:

- The system should work seamlessly across web, Android, and iOS.
- Deployment is expected in **low-latency environments (HQ)** and **high-latency/no-internet areas (field)**.
- Expected to serve **50M+ global users** during peak disaster situations.
- Timeline: Go live in **6 months** for the upcoming monsoon season in Asia.

- Budget is **limited**, but failure would lead to loss of lives, reputational damage, and loss of funding.
- Stakeholders include: field volunteers, donors, governments, media, local NGOs, data scientists, and end-beneficiaries.
- Everything must be **highly secure**, but users in the field cannot remember complex passwords or 2FA.
- They expect **advanced features** like offline maps, real-time collaboration, analytics — but the team is short-staffed and some parts may be outsourced.

Questions

- What should you prioritize in the system design?
- What potential conflicts or contradictions do you notice?
- What areas feel the most confusing or hard to decide?